

Autonomous Robotic Localization & AI Workflows

Scalable Edge Computing and Mathematical Modeling

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Presentation Agenda

- 1 Motivation & Challenges
- 2 Methodology & Architecture
- 3 Experimental Results
- 4 Conclusion

Problem Formulation: GPS-Denied Inspection

Key Operational Bottlenecks:

- Extreme wheel slippage and track deadband in industrial environments.
- Severe non-uniform velocity profiles across repetitive traversal runs.
- Strict computational and memory budgets on embedded edge devices ($< 5\text{ W}$).

Core Objective

Develop a unified spatial-temporal alignment engine achieving sub-centimeter trajectory tracking without cloud dependencies.

Swept Circular Arc Kinematics

Discrete Pose Integration

Given displacement increments Δs_L and Δs_R over track gauge W :

$$\Delta\theta = \frac{\Delta s_R - \Delta s_L}{W}, \quad \Delta s = \frac{\Delta s_L + \Delta s_R}{2} \quad (1)$$

- **Linear Displacement:** $x_{k+1} = x_k + \Delta s \cdot \cos(\theta_k + \Delta\theta/2)$
- **Lateral Displacement:** $y_{k+1} = y_k + \Delta s \cdot \sin(\theta_k + \Delta\theta/2)$

Benchmark Evaluation

Method	RMSE Error (m)	Max Drift (m)	Latency (ms)
Raw Encoder Dead-Reckoning	0.384	0.812	0.4
Standard Particle Filter	0.092	0.184	18.2
Ours (Constrained DTW + Arc)	0.012	0.024	4.1

Key Takeaway

Achieved a **76% error reduction** while operating within a real-time 100 Hz execution envelope.

- Presented a lightweight, production-grade localization and document workflow.
- 100% reproducible, open-source, and offline capable.

Thank You!

Questions & Discussion • alex.morgan@email.com